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Thermohaline structure and mixing in the region of Prydz Bay, Antarctica

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Abstract—Temperature and salinity data, collected during successive summer expeditions to the region of Prydz Bay, Antarctica between 1981 and 1985, indicate that the mixing processes responsible for the thermohaline circulation are similar to those in the Weddell Sea. In particular, Circumpolar Deep Water intrudes over the continental shelf break, aided by periodic upwelling processes associated with the tides and continental shelf waves, and mixes with the colder Shelf Water associated with the formation of sea ice over the continental shelf. Mixing of these intrusive water masses is enhanced by shear, and the resultant cold and salty mixture, termed here Prydz Bay Bottom Water, flows westward and downslope under the influence of the Coriolis force and Antarctic coastal winds. Prydz Bay Bottom Water is insufficiently dense to reach abyssal depths, and probably interleaves with the Circumpolar Deep Water at intermediate depths. The mixing processes are likely to be responsible for active bottom water formation in Prydz Bay for most of the year, except perhaps in the summer months when the mixture is insufficiently dense to flow down the continental slope.

INTRODUCTION

THE continental shelves of Antarctica are among the most active regions of the world in terms of large-scale thermohaline circulation. MOSBY (1934) showed that if continental shelf waters became sufficiently salty as a result of sea-ice formation, they could sink down the continental slope by virtue of their greater density to form bottom water. The mixing processes involved are complex, and the majority of work associated with obtaining an understanding of these mixing processes has been undertaken in the Weddell Sea, the principal source region for Antarctic Bottom Water (e.g. FOSTER *et al.*, 1987).

By contrast, the level of activity on the Antipodean side of the continent has been substantially lower, although early work by GORDON and TCHERNIA (1972) and GORDON (1972) has shown that the mixing processes appear to be similar to those operating in the Weddell Sea. More recent work by SMITH *et al.* (1984) discussed data collected during FIBEX (First International Biomass Experiment) by the Australian Antarctic Division during the summer of 1980–1981 in the region of Prydz Bay, between 60° and 90°E.

This paper describes and interprets data comprised essentially of vertical profiles of conductivity, temperature and depth collected in the Prydz Bay region (50°–90°E and 60°–70°S) in the summers of 1982–1985 on four Australian Antarctic Research Expedition cruises. By comparing the hydrological data with those collected in the more

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intensively studied Weddell Sea, we can draw general conclusions about the general circulation and its relation to bottom and deep water formation. The cruises were named FIBEX (January–February 1981), ADBEXI (Antarctic Division Biomass Experiment, Phase 1, November–December 1982), ADBEX2 (Phase 2, January 1984), SIBEX2 (Second International Biomass Experiment, February–March 1985), and ADBEX3 (Phase 3, October–November 1985).

THE OBSERVATIONAL PROGRAM

A bathymetric map of the Prydz Bay region is shown in Fig. 1a, while the locations of selected stations from which data was drawn for this paper are shown in Fig. 1b. The sampling location and depth of each station was determined by the needs of the biological programs with the result that the majority of the stations were in the deep ocean, well off the continental shelf. ADBEX1 was an exceptional cruise having stations mainly on the shelf of Prydz Bay proper, between 70° and 78°E. Few stations were located on the continental slope so that the transition from deep ocean to shelf waters was not measured in detail. Few deep ocean stations were sampled to the bottom so that sections are shown to 600 or 1000 m only. In addition, sampling was not geographically systematic and only during SIBEX2 were stations sampled on a grid basis. ADBEX3 was the earliest season cruise and included the most westerly stations, sampled while M.V. *Nella Dan* was beset in ice for 2 months. The data were collected with a Neil Brown Mark III CTD, and were not supported by independent evaluations of temperature and salinity from water samples and thermometer measurements. The CTD was calibrated before and after each cruise by the manufacturer, resulting in accuracy better than 0.005°C in temperature and 0.01 in salinity.

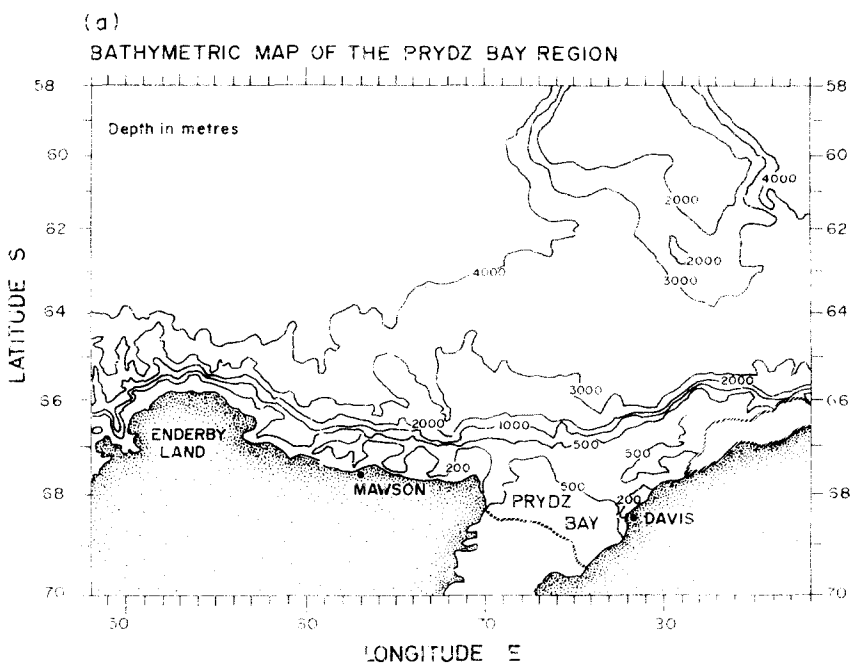


Fig. 1a. The bathymetry of the Prydz Bay region of Antarctica (after GEBCO chart 5–18).

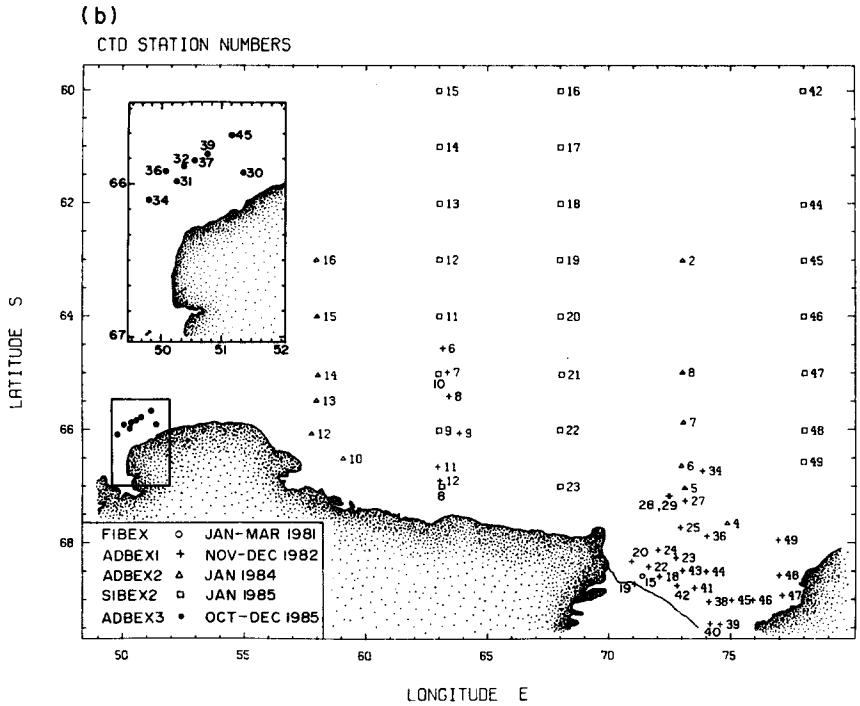


Fig. 1b. Positions of stations referred to in this paper at which conductivity–temperature–depth profiles were measured during Antarctic Division cruises from 1981 to 1985.

THE GENERAL CIRCULATION

Satellite passive-microwave observations of Antarctic Sea Ice by ZWALLY *et al.* (1983) show that mean monthly sea-ice concentrations (obtained from a 4-year average) are generally much greater than 70% from April to December in Prydz Bay, with open water predominant essentially only in January. Air temperatures are generally below freezing even in summer, and freezing processes are active for most of the year.

An analysis of monthly mean wind stress by HAN and LEE (1981) shows that the average wind stress in the Austral summer is primarily zonal, with a west-going component near the continent, and an east-going component at about 60°S. From November to January the coastal winds produce an average wind stress of about 0.1 N m^{-2}, however, this increases to over 0.2 N m^{-2} by March. The well-known (DEACON, 1937) features of a west-going coastal current and an east-going deep ocean current farther offshore with a divergence zone separating them are fully consistent with this forcing. TCHERNIA and JEANNIN (1984) present iceberg tracks illustrating westward drift along the continental shelf with stronger eastward drift further north in the Antarctic Circumpolar Current, while drift tracks of ICEX buoys (ALLISON, 1985) also follow the same trends.

From the present data, the larger scale baroclinic circulation is also consistent with this picture. An idea of the baroclinic circulation in Prydz Bay can be obtained from contours of geopotential anomaly from selected ADBEX1 stations (Fig. 2a). Only those stations having data to depths greater than 500 m were used to draw contours, ensuring a

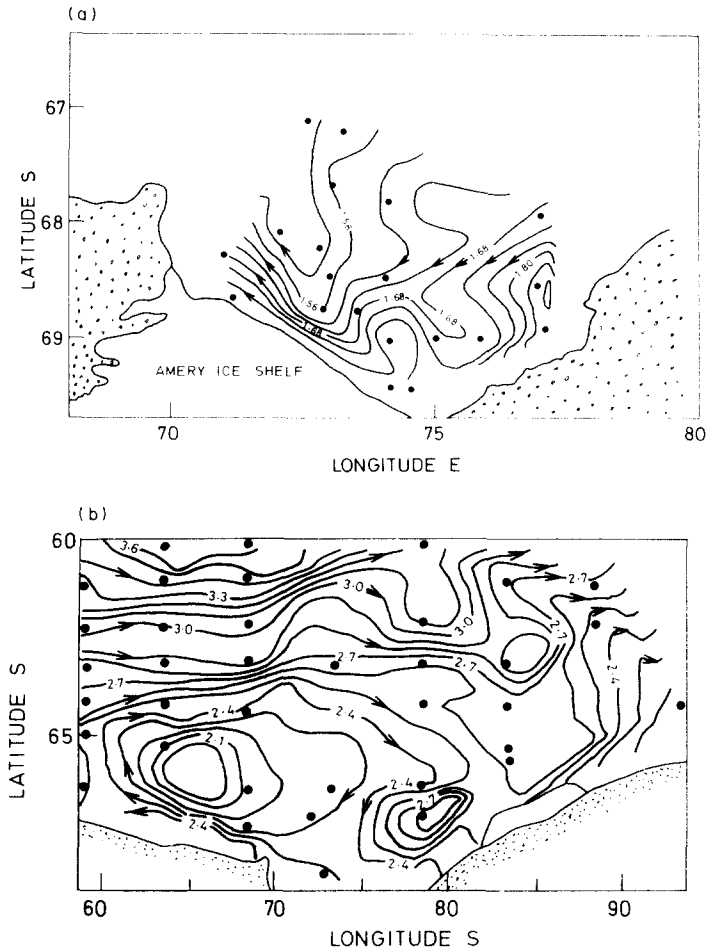


Fig. 2. (a) Contours of geopotential anomaly ($m^2 s^{-2}$) drawn relative to 500 m for the Prydz Bay region from ADBEX1 data. Dots indicate the positions of stations used to contour the data. (b) Contours of geopotential ($m^2 s^{-2}$) drawn relative to 600 m for the Southern Ocean region north of Prydz Bay from SIBEX data. Dots indicate the positions of stations used to contour the data.

reasonable geographic spread of anomalies. Calculations of geopotential anomaly were then made relative to 500 m. The contours show a diffuse inflow into Prydz Bay from the east, and a stronger coastally confined outflow to the west. These features might be expected for a flow that obeys conservation of potential vorticity by following depth contours. Figure 2b shows the geopotential anomaly contours relative to 600 m for a larger region surrounding Prydz Bay. To the north, the contours show a positive northward gradient, consistent with the eastward-flowing Antarctic Circumpolar Current; however, the coastal regions show strong variability with occasional east-flowing currents in addition to the wind-driven west-flowing currents. The position and structure of the zone of divergence also are clearly highly variable. It is important to remember

that the data are not synoptic, so the contours are only an approximate representation of the flow at one period of time.

STRUCTURE OF THE WATER COLUMN

Surface waters in Antarctic Oceans are highly variable owing to the relative strengths of the freezing and melting influences, however; the summer surface layers rarely extend beyond 60 m depth. Below, extending to perhaps 200 m in the open ocean, lies the Winter Water (WW) layer whose properties are a result of wintertime convection with temperature $T < -1.5°C$, and salinity S in the range 34.2–34.56 (Figs 3 and 4). In the surface layers of the open ocean, salinity increases due to salt exclusion from sea-ice formation are restricted because of convective mixing with less saline waters from below. By contrast, a saltier and denser water mass (Shelf Water, or SW) forms over the continental shelf when the sea surface is being actively frozen. SW is characterized by $T < -1.5°C$, and $S > 34.56$. Offshore but at intermediate depths lies Circumpolar Deep Water (CDW) having T from 0.5 to 2.0°C, and S from 34.50 to 34.75 (Carmack, 1977).

In the Weddell Sea, the major source region for Antarctic Bottom Water, salinities may reach 34.70. Also, the warm deep water of the Weddell Sea is generally cooler than CDW and rarely exceeds 1°C; it is usually referred to simply as Warm Deep Water (WDW), having $0 < T < 1°C$ but salinities similar to that of CDW. Deep convection in the Weddell Sea is a result of complex mixing processes (Foster and Carmack, 1976). Modified Warm Deep Water (MWDW), a mixture of WDW and WW, intrudes over the continental shelf and mixes with the SW in the region of the continental shelf break. The resultant mixture (Weddell Sea Bottom Water, or WSBW) flows off the continental shelf and down the slope as a consequence of having a density greater than that of the WDW. The dense mixture entrains the WDW as it descends, and the resultant water mass is termed Antarctic Bottom Water (AABW). One of the objectives of this paper is to compare mixing processes in Prydz Bay with those in the Weddell Sea.

The observed temperature structure from ADBEX1 and ADBEX2 (Fig. 3a,b) show the cold SW water masses inshore of the shelf break, with strong horizontal temperature gradients occurring in the vicinity of the shelf break. Reference to individual station profiles shows that these strong gradients are associated with numerous intrusions. Salinities over the shelf (not shown here) are generally less than 34.50 with a general increase of salinity with depth throughout the water column, suggesting that the cooler waters are a remnant of winter convection rather than the result of active freezing processes. The doming of the isotherms at Stas 2 and 8 (Fig. 3b) from ADBEX2 is indicative of the divergence zone. Potential temperature sections from SIBEX2 (Fig. 4a,b) also show the cooler temperatures over the shelf proper; however, in each case, the shelf is quite narrow, and the cooler water masses extend farther offshore than the shelf break. Maximum salinities, which occur near the bottom, are: 34.52 (ADBEX2, Sta. 4), 34.58 (SIBEX2, Stas 23 and 49), and 34.61 (ADBEX1, Sta. 44). The salinities from ADBEX1 are sufficiently high that the SW is slightly denser than the adjacent WDW; however, from the temperature sections there is no evidence of active convection down the continental slope.

Substantially more information can be obtained from temperature–salinity plots of all data from the cruises. These are shown in Fig. 5, with one dot plotted every 10 m of depth for all stations. The data from the top 50 m does not appear here, since they are

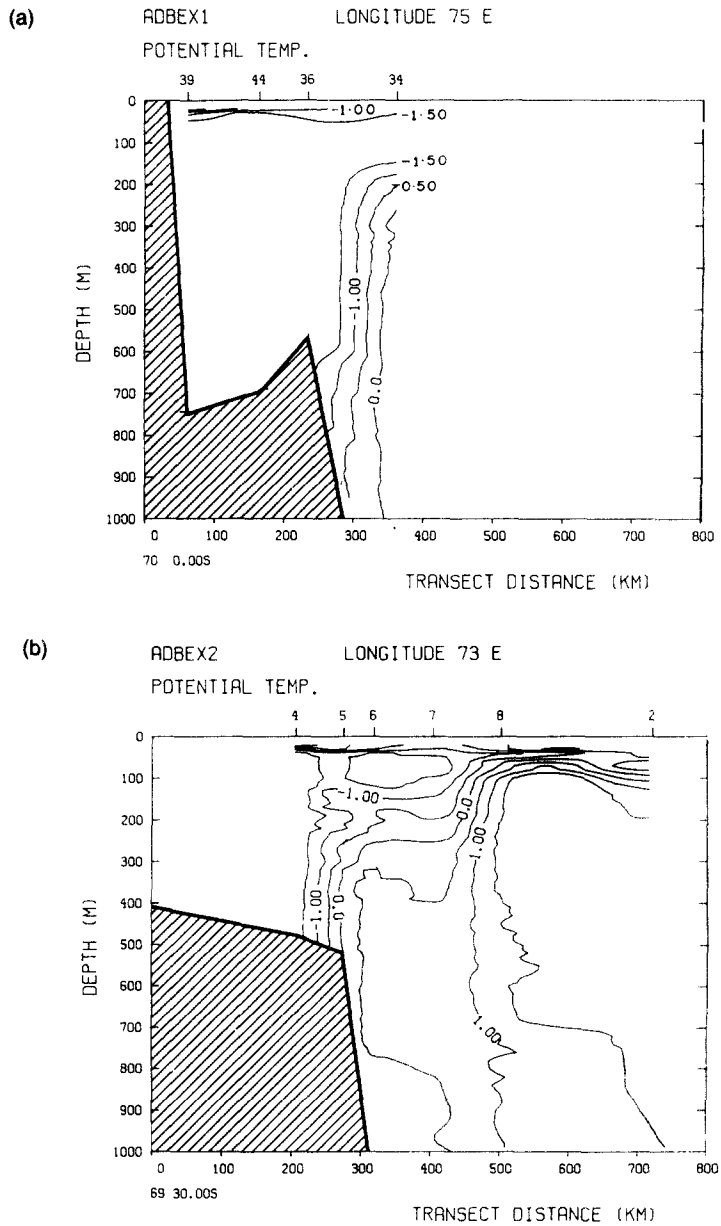


Fig. 3. Potential temperature sections at (a) 75°E drawn from ADBEX1 data, and (b) 73°E drawn from ADBEX2 data.

highly variable. The distribution of data on the T–S diagrams appears to be similar for all cruises, but there are some important differences. In particular, only the ADBEX1 data from within Prydz Bay proper show SW with S > 34.56. In this case, deep convection down the slope could lead to the formation of deep interleaved layers. Indeed, salinities are occasionally greater than 34.62, suggesting that the SW is sufficiently dense that it may flow down the continental slope under the adjacent CDW to form true AABW. The

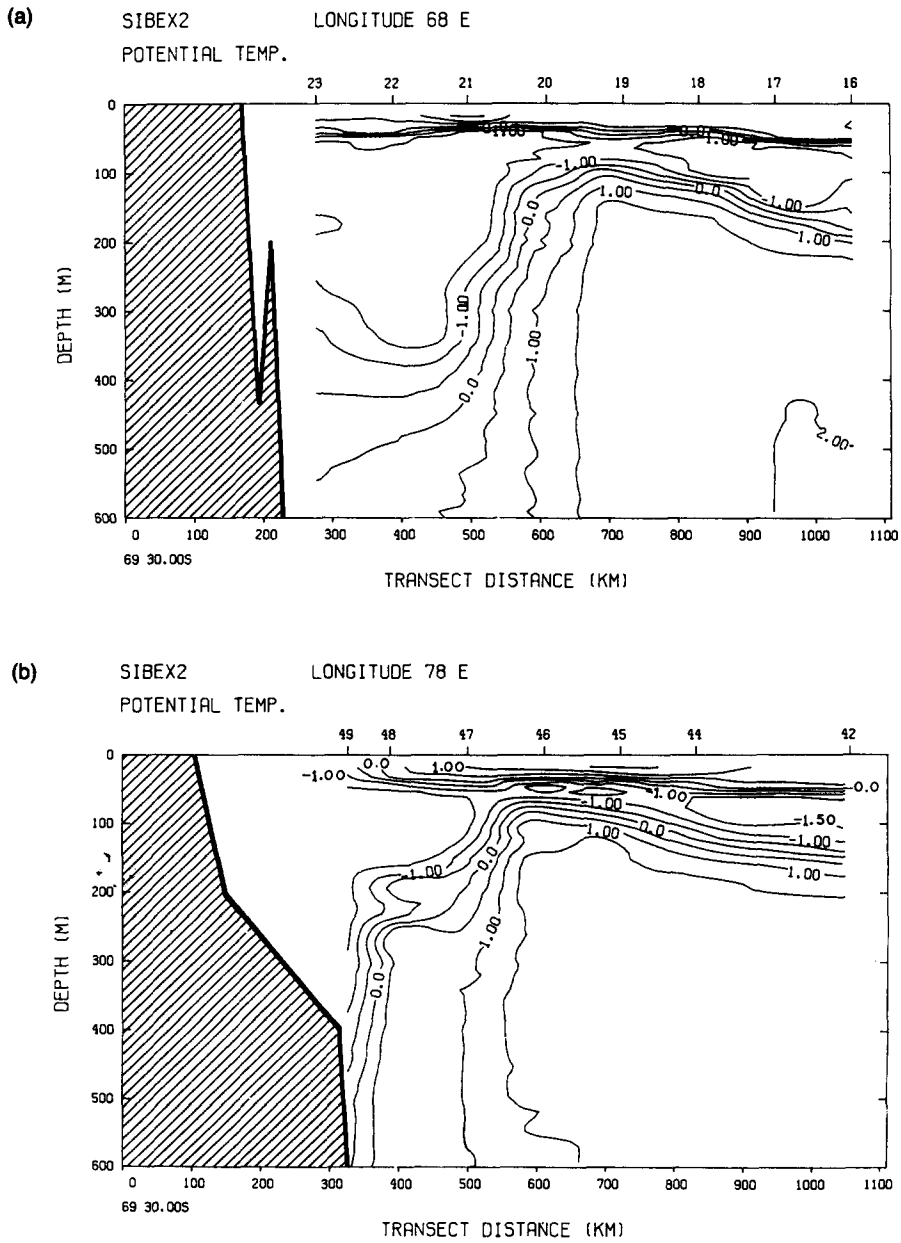
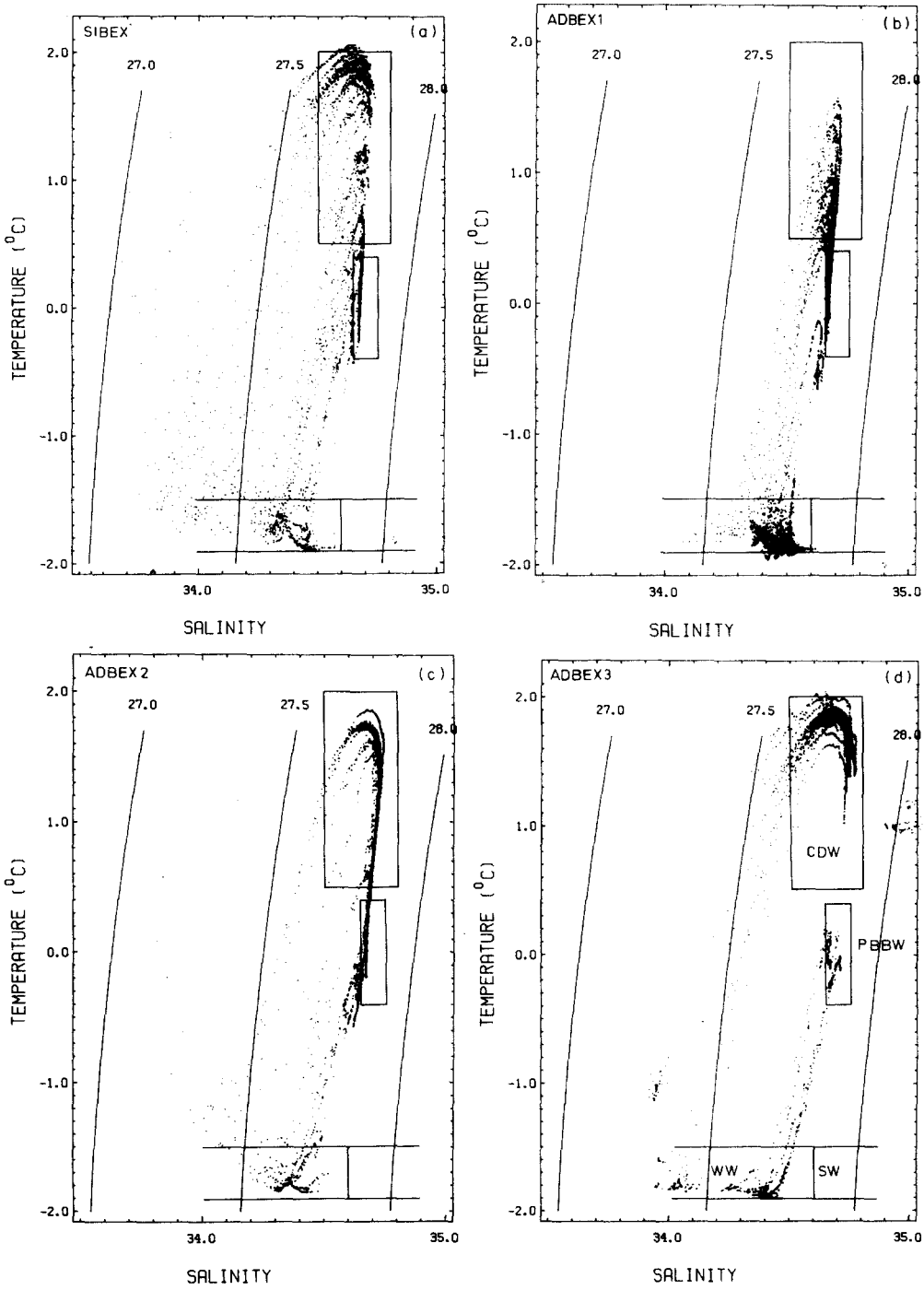


Fig. 4. Potential temperature sections at (a) 68°E and (b) 78°E drawn from SIBEX2 data.

data were obtained early in the season (November–December) when active surface freezing was occurring. As described above, however, there is no indication of flow directly downslope off the shelf adjacent to Prydz Bay.

The explanation for this apparent inconsistency is clear once we consider the wind data, since the west-going (easterly) winds near the continent will drive the continental shelf waters primarily alongshore to the west in a manner analagous to that which occurs


SIBEX
(a)
ADBEX1
(b)
ADBEX2
(c)
ADBEX3
(d)
27.0
27.5
28.0
TEMPERATURE (°C)
SALINITY
2.0
1.0
0.0
-1.0
-2.0
34.0
35.0
CDW
PBBW
WW
SW

in the Weddell Sea. Any downslope flow therefore would be expected west of Prydz Bay, and would result in an alongshore flow in approximate geostrophic balance having only a small downslope component (FOSTER *et al.*, 1987). ADBEX1 Sta. 11 had relatively dense water with properties of $T < 0°C$ and $S > 34.62$ at depths of 500–1500 m on the continental slope, lending support to this hypothesis. Reference to the T–S diagram for ADBEX3 (Fig. 5d) shows no high-salinity SW, but does show a mixture with anomalous properties centred near $T = 0°C$ and $S = 34.65$ (Stas 31, 37, 39 and 45). This mixture is found on the continental slope, at depths between 700 and 1600 m, where CDW with temperature exceeding 0.5°C usually occurs away from the continental shelf ($T < 0°C$ usually occurs at depths greater than about 2000 m, or in the surface 100 m). This anomalous water mass, however, does have properties consistent with a mixture of CDW and highly saline SW (probably from Prydz Bay). This highly saline SW could not have been observed during ADBEX3 because there were no profiles taken in Prydz Bay proper that season, but the existence of this highly saline SW seems highly probable since it was clearly evident in Prydz Bay during ADBEX1. Also, the density of some of this anomalous mixture is slightly higher than that of any other water mass observed during these cruises (with the exception of those in Prydz Bay proper), adding further weight to the hypothesis that the mixture is a mixture of highly saline SW from Prydz Bay, and WDW. It may therefore be called Prydz Bay Bottom Water (PBBW). It is important to note that, although waters having similar temperature and salinity properties were observed during SIBEX, ADBEX1 and ADBEX2, these waters were at depths exceeding 1500 m and therefore were not indicative of active convection.

Thus it appears that the wider shelf in Prydz Bay allows a greater build-up of high-salinity SW, and the wind stress drives the SW to the west. The combined effects of greater density and narrowing of the shelf to the west tend to drive the SW to the shelf edge where, because of its greater density, it gains a small downslope component. Admixture with the CDW results in a mixture (PBBW) having properties half way between those of SW and WDW. This mixture is insufficiently dense, compared to Weddell Sea Antarctic Bottom Water, to reach abyssal depths and so it probably interleaves at some intermediate depth as described by CARMACK and KILLWORTH (1978) for the Adelie coast and FOSTER and MIDDLETON (1980) for the Weddell Sea. This mixing process seems somewhat simpler than that of the Weddell Sea, since the role of the WW layer (via MWDW) appears to be unimportant here. This may be a result of the CDW waters being essentially warmer than the WDW of the Weddell Sea; however, it would require a data set obtained by following the path of the Prydz Bay outflow to delineate further the details of the mixing processes.

The principal features of circulation described here are supported by current and temperature data acquired during 1985 from three moorings in the Prydz Bay region and

Fig. 5. (a) Temperature–salinity diagrams for all data collected during Antarctic Division expedition SIBEX2. CDT stations were usually taken to 600 m depth where there was sufficient depth, except for four stations at 66°S having longitudes of 58°E, 63°E, 68°E and 73°E. (b) Temperature–salinity diagrams for all data collected during Antarctic Division expedition ADBEX1. CTD stations were generally taken to 1000 m depth where topography allowed. (c) Temperature–salinity diagrams for all data collected during Antarctic Division expedition ADBEX2. CTD stations were generally taken to depths of 1000 m where topography allowed. (d) Temperature–salinity diagrams for all data collected during Antarctic Division expedition ADBEX3. CTD stations were generally taken to depths of 1000 m where topography allowed. Water masses labelled CDW (Circumpolar Deep Water), WW (Winter Water), SW (Shelf Water) and PBBW (Prydz Bay Bottom Water) are described in the text.

described by HODGKINSON *et al.* (1988). The plots presented in the report confirm the existence of near freezing temperatures within Prydz Day throughout most of the year, with pulses of warmer water intruding over the continental shelf break at diurnal periods throughout the year. Progressive vector diagrams drawn from the data acquired at mooring 4 (in 630 m water depth directly offshore from Mawson Station) confirm a somewhat larger downslope component of near freezing water toward the bottom, consistent with the flow of denser water off the shelf as discussed above.

SHELF DYNAMICS AND MIXING PROCESSES

FOSTER *et al.* (1987) have identified the combined effects of sub-inertial frequency motions and semidiurnal tides in the mixing processes on the shelf of the southern Weddell Sea. In particular, they argue that it is the barotropic sub-inertial frequency continental shelf waves that upwell WDW periodically onto the shelf, while mixing of the cold SW with the WDW is promoted through stirring induced by shear instabilities and vertical gradients associated with the vertical shear in the semidiurnal tides.

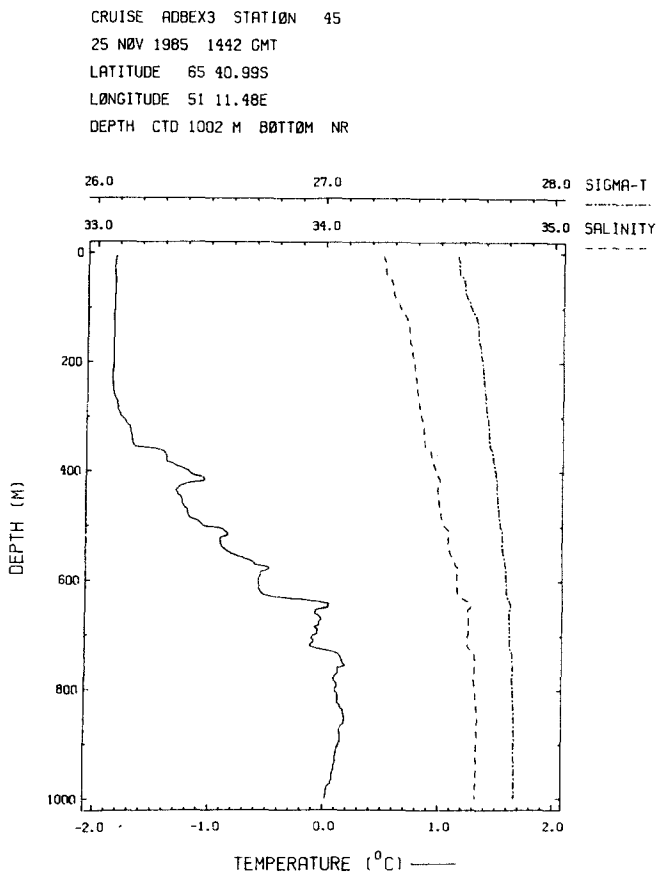


Fig. 6. Temperature, salinity and density profiles for ADBEX3 Sta. 45, showing numerous intrusive layers between 300 and 800 m depth.

The coastal winds in the Prydz Bay region are often fierce and highly variable (STRETEN, 1968). This variability provides the generation mechanism for continental shelf waves, which manifest themselves as periodic current pulses propagating to the west on Antarctic shelves (MIDDLETON *et al.*, 1982, 1987). The pulses become superimposed on the mean current driven by the easterly winds.

On the continental shelf near Prydz Bay, these continental shelf waves will provide a general periodic upwelling of CDW onto the shelf, allowing the tides to enhance mixing through vertical shearing of intrusions. Intrusive layers are commonly seen near the shelf break regions as exemplified by ADBEX3 Sta. 45 (Fig. 6). Diurnal tides also will propagate along the coast as continental shelf waves at these latitudes, and the tidal currents measured at the shelf break are mostly diurnal (HODGKINSON *et al.*, 1987). Since there seems to be barotropic energy at both diurnal and lower frequencies at the shelf break, it appears that the diurnal tides are responsible for the periodic upwelling of CDW onto the continental shelf, at least in Prydz Bay. Farther west, the stronger currents are of much lower frequency (HODGKINSON *et al.*, 1987), and it appears that diurnal frequency variability will be less important for upwelling than lower frequency variability.

Although the southern Weddell Sea shelf is somewhat farther south (75°S) than Prydz Bay (67°S), the salient features of a wind-driven circulation on a wide shelf subject to surface freezing through much of the year are common, resulting in the production of salty, cold and dense SW over the continental shelf. As a result of the Coriolis force and the westward wind-driven circulation, and after mixing the CDW, the dense SW appears as PBBW on the slope at substantial distances farther west, with a small downslope velocity component. Deep convection from Prydz Bay appears to be less effective than that in the Weddell Sea, probably because of the narrower shelf, and warmer adjacent CDW; however, similar mixing and thermohaline circulation mechanisms are operating in both regions.

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